The S-Curve After the Shock: Machine Learning Evidence on Mortgage Prepayment in the Lock-In Era

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Abstract

We estimate loan-level mortgage prepayment hazard models—logistic regression, gradient-boosted trees, and a deep neural network in the spirit of Sadhwani et al. (2021)—on 18.5 million Fannie Mae loans observed monthly between 2018 and 2025, and use them to characterize how the prepayment function changed after the 2022 interest-rate shock. We report three findings. First, the regime break was costly in both directions: a model estimated through 2021 over-predicts aggregate speeds by approximately 9 CPR points at the onset of the shock, yet annual re-estimation *worsens* 2023 forecasts relative to the frozen model (3.8 versus 2.3 CPR points of UPB-weighted cohort error), because re-estimation on the brief transition period imports an upwardly biased speed level. Re-estimation dominates only from 2025 onward. Second, partial-dependence analysis of the learned prepayment function on a common reference population shows that the lock-in era did not flatten the refinancing S-curve but rotated it: the refinancing response above the at-the-money point is substantially steeper than in the pre-pandemic regime (30.5 versus 13.2 CPR at +150 basis points of incentive), while deep-discount turnover operates primarily through the prevailing-rate level. Third, applying the estimated model to the December 2025 cross-section, the 2023–24 high-coupon universe exhibits pronounced negative convexity: projected speeds for 7.0% coupons rise from approximately 25 CPR at unchanged rates to above 50 CPR under a 150 basis point rally. We document three data-quality pitfalls in the public dataset that mechanically inflate out-of-sample classification metrics if untreated, and we release all code and artifacts for replication.

Keywords: mortgage prepayment; MBS; machine learning; lock-in effect; gradient boosting; interpretability.

1 Introduction

Between March 2022 and October 2023 the 30-year fixed mortgage rate in the United States rose from below 4% to nearly 8%, the fastest repricing of the mortgage market in four decades.

*This paper is a personal research project based exclusively on publicly available data. The views expressed are the author's own and do not represent those of any employer. Code and replication materials are available in the accompanying public repository. Not investment advice.

Aggregate conditional prepayment rates (CPR) on Fannie Mae collateral fell from above 35 to below 5, and by the end of 2023 nearly the entire agency universe was out of the money. The episode placed prepayment models—the central pricing input for a $9 trillion market—outside the support of their training data. The practitioner record of the aftermath is extensive: vendor models missed monthly aggregate prints by 10–25% through 2023–2025, disagreed on the sign of convexity below par, and generated relative-value signals that inverted against realized returns (Yu, 2023; Katz and Li, 2024).

A parallel academic literature has quantified the underlying mechanism—rate lock-in—from the housing side. Batzer et al. (2024) estimate that each percentage point by which prevailing rates exceed a borrower's contract rate reduces the probability of sale by 18.1%; Fonseca and Liu (2024) find a 9–16% reduction in moving probability per percentage point of lock-in; Liebersohn and Rothstein (2025) attribute a 16% decline in the mobility of mortgaged households to the mechanism. This literature measures the turnover component of prepayment, but does not address the model-side question that concerns fixed-income practitioners: *what exactly changed in the prepayment function, and how should data-driven models be managed across such a break?*

This paper provides one of the first public model-side treatments of the 2022 regime break estimated on open data. We construct a monthly discrete-time hazard panel from the Fannie Mae Single-Family Loan Performance dataset (18.5 million loans, 2018–2025), estimate three classes of prepayment model of increasing flexibility under a strict walk-forward protocol, and evaluate them at two levels: loan-level discrimination and—the economically relevant metric—UPB-weighted cohort-level CPR error against full-population actuals.

Our contributions are threefold. First, we quantify the *re-estimation trap*: while the pre-shock model fails immediately and visibly (Section 5.2), naive annual re-estimation underperforms the frozen legacy model for roughly two years after the break, because parameters fitted to the brief transition sample import a biased level. This finding has direct implications for model governance and is consistent with, but sharper than, the documented whipsaw of commercial models. Second, using per-regime estimation and partial-dependence analysis on a fixed reference population (Section 5.3), we decompose the apparent post-2022 flattening of the S-curve into composition and behavior. Conditional on composition, the refinancing limb of the lock-in-era S-curve is *steeper* than its pre-pandemic counterpart—the refinancing elasticity through the cusp rises from 2.2 to 12.6 CPR points per 100 basis points—while depressed deep-discount turnover is absorbed by the prevailing-rate level rather than by a decayed incentive response. This decomposition speaks directly to a live practitioner debate (Landy, 2024; Katz and Li, 2024). Third, we translate the estimates into a forward-looking statement about the current coupon stack (Section 5.4): as of December 2025, model-implied speeds for the 2023–24 vintage 6.5–7.0% coupons rise steeply under modest rallies, tripling for 6.5s between unchanged rates and a 150 basis point rally.

A secondary, methodological contribution concerns data quality. Three fields in the public dataset are blanked on a loan's terminal record, so that—if used contemporaneously—their missingness perfectly reveals the prepayment event. Untreated, these produce out-of-sample AUCs of 0.9999 and would invalidate any study built on them. Section 3.4 documents the mechanism and the corrections, which we regression-test in the released code.

2 Related literature

Machine learning for mortgage risk. Sadhwani et al. (2021) estimate multi-state transition models on 120 million U.S. mortgages with deep neural networks and document large out-of-sample gains over logistic benchmarks, with prepayment exhibiting the strongest nonlinearities. Zhang et al. (2019) deploy neural networks for agency MBS prepayment at pool level and interpret the fitted model through implied S-curves, burnout, and seasonality; Schultz and Fabozzi (2021) make the case for gradient-boosted trees at loan level. Blumenstock et al. (2022) find machine-learning survival models dominate statistical benchmarks for competing default and prepayment risks; Ojha and Lee (2021) compare conventional and deep learners on Fannie Mae data through the financial crisis. Relative to this literature, our contribution is not architectural novelty but the regime question: all published samples of which we are aware end before or shortly after the 2022 break, and none addresses model management across it.

The lock-in effect. Batzer et al. (2024), Fonseca and Liu (2024), and Liebersohn and Rothstein (2025) establish the magnitude of rate lock-in for housing turnover, labor mobility, and welfare. We use their elasticities as external anchors for the turnover-suppression channel recovered by our models.

Practitioner record. Yu (2023) frames post-2022 prepayment modeling as an out-of-sample problem; monthly commentary by Santander US Capital Markets documents production model errors of 10–25% through 2023–25 and argues that apparent S-curve flattening reflects spread-at-origination (SATO) composition (Landy, 2024); Katz and Li (2024) document model-implied relative value inverting against realized returns in 2024 and, in early 2026, pool-level S-curves exceeding pandemic-era steepness. Our per-regime, composition-controlled S-curves reconcile these observations within a single estimated model.

3 Data

3.1 Loan-level performance data

We use the Fannie Mae Single-Family Loan Performance dataset, acquisition quarters 2018Q1 through 2025Q4: 18,468,917 fixed-rate loans and their complete monthly servicing records through December 2025, parsed from the raw 113-field pipe-delimited files. The window brackets three regimes: pre-pandemic (2018–19), the pandemic refinancing wave (2020–21), and the lock-in era (2022–25). Voluntary prepayment is identified by zero-balance code 01 (payoff); credit events and repurchases are treated as censoring.

3.2 Cohort-level ground truth

From the full population—prior to any sampling—we compute actual cohort-level single monthly mortality and CPR by vintage year and coupon (rounded to the nearest 0.5), where SMM is voluntarily prepaid UPB over beginning-of-month UPB. These series serve as the evaluation target for all models and align with public anchors: 2023-vintage 7.0 coupons print 5.4 CPR in June 2024, peak at 30.3 in the October 2024 refinancing episode, and revert to 9.1 by December (Federal Housing Finance Agency, 2024); 2021-vintage 2.5 coupons remain at 2–4 CPR throughout 2023–25; 2018-vintage 4.5 coupons run at 41–57 CPR through the pandemic wave (Haughwout et al., 2023).

3.3 Estimation panel

A deterministic, hash-based stratified sample (vintage year $\times$ note-rate bucket, with inverse-probability loan weights) of 1,201,243 loans yields a monthly hazard panel of 2,532,987 loan-month observations containing all 428,793 voluntary prepayment events and a 5% importance-weighted subsample of non-event months. Weights are carried through estimation and aggregation, so fitted hazards are on the monthly SMM scale and cohort aggregates are population-representative.

Covariates comprise: the refinancing incentive (note rate minus the Freddie Mac PMMS 30-year survey rate); spread at origination (SATO); burnout (cumulative positive incentive over prior months); mark-to-market LTV (original LTV revalued with state-level FHFA house-price indices); loan age; original balance; credit score; DTI; original LTV; calendar month; the prevailing mortgage-rate level; and categorical origination attributes (channel, purpose, property type, occupancy, state, borrower count, first-time-buyer flag) plus lagged delinquency status and modification flag.

3.4 Removal-record measurement pitfalls

Three covariates initially produced out-of-sample AUCs of 0.9999—a value that should be read as diagnostic of leakage rather than of predictive skill. The mechanism is systematic: the dataset blanks monthly-reported fields on a loan's terminal record. Consequently (i) mark-to-market LTV computed on the end-of-period balance equals exactly zero if and only if the loan prepaid that month; (ii) the modification flag and delinquency status are missing on removal records, so their missingness indicators reveal the event; and (iii) the reported loan-age field is likewise blanked. All state-dependent inputs are therefore measured as of the *beginning* of the month—balances lagged (the factor-date convention), status fields lagged one month, and loan age computed from calendar arithmetic rather than read from the file. Each correction carries a regression test in the released code. Post-correction discrimination falls to the 0.59–0.75 range, which we regard as the honest difficulty of monthly voluntary-prepayment classification.

4 Methodology

4.1 Hazard framework and inverse-probability weighting

Let $y_{it} \in \{0, 1\}$ indicate voluntary prepayment of loan i in month t. We model the discrete-time hazard $h_{it} = \Pr(y_{it} = 1 \mid x_{it})$, where x_{it} contains only information available at the beginning of month t.

The estimation panel is drawn from the population in two independent, deterministic stages, and every observation carries the inverse of its inclusion probability. Stage one samples *loans*:

within stratum $s(i)$ (vintage year $\times$ note-rate bucket), loan i is included with probability $\pi_{s(i)} = \min\{1,\, \bar{n}/N_{s(i)}\}$, where N_s is the stratum size and $\bar{n}$ the per-stratum target, implemented by a hash of the loan identifier (no random-number generator, hence exactly replicable). Stage two samples *loan-months*: every event month ($y_{it} = 1$) of a sampled loan is retained with probability one, while non-event months are retained with probability $\kappa = 0.05$, again by hashing. The combined design weight is therefore

$$w_{it} \;=\; \begin{cases} 1/\pi_{s(i)}, & y_{it} = 1, \\ 1/\big(\pi_{s(i)}\,\kappa\big), & y_{it} = 0. \end{cases} \tag{1}$$

For any population functional expressible as a sum over loan-months, $S = \sum_{it} z_{it}$, the weighted sample sum $\hat{S} = \sum_{it \in \mathcal{S}} w_{it} z_{it}$ is a Horvitz–Thompson estimator and is unbiased, since each term's inclusion probability is exactly $\pi_{s(i)}$ or $\pi_{s(i)}\kappa$ by construction. Two consequences follow. First, maximizing the w-weighted log-likelihood targets the population likelihood, so fitted hazards are calibrated to the population base rate—retaining all events while down-sampling non-events does *not* require the usual post-hoc intercept correction for case-control sampling, because the weights already undo the design. Second, population cohort-level speeds are recovered by weighting fitted hazards with the product of design weight and beginning-of-month balance b_{it}: for cohort c in month t,

$$\widehat{\mathrm{SMM}}_{ct} \;=\; \frac{\sum_{i \in c} w_{it}\, b_{it}\, \hat{h}_{it}}{\sum_{i \in c} w_{it}\, b_{it}}, \qquad \widehat{\mathrm{CPR}}_{ct} \;=\; 1 - \big(1 - \widehat{\mathrm{SMM}}_{ct}\big)^{12}, \tag{2}$$

where numerator and denominator are each Horvitz–Thompson estimators of their population counterparts (expected voluntarily-prepaid balance and total beginning balance). Evaluation never relies on the sampled denominator where the population value is available: actual cohort CPRs are computed on the *full* population of Section 3.2. Appendix A verifies empirically that results are insensitive to halving κ, and that fitted hazards are well calibrated against weighted empirical event rates.

4.2 Estimators

We estimate three models of increasing flexibility on identical features and weights: (i) weighted logistic regression (median-imputed numerics, one-hot categoricals)—the linear benchmark; (ii) gradient-boosted trees (LightGBM; 63 leaves, 600 trees, learning rate 0.05, minimum 100 observations per leaf), with missing values handled natively; and (iii) a feed-forward neural network with categorical embeddings (three hidden layers, 128–128–64 ReLU units, dropout 0.1), trained with weighted cross-entropy, following the architecture spirit of Sadhwani et al. (2021). We deliberately avoid early stopping on trailing-window validation sets: in the presence of regime shifts, a trailing window is unrepresentative of both past and future, and we found it induces severe underfitting precisely at the breaks.

4.3 Evaluation protocol

All evaluation is strictly walk-forward: models are estimated on months up to December of year T and evaluated on the following twelve months, for $T = 2018, \ldots, 2024$. We report loan-level weighted AUC, and UPB-weighted mean absolute cohort CPR error (in CPR points) against the full-population actuals of Section 3.2. We emphasize the latter: loan-level discrimination metrics have limited economic meaning for portfolio speeds, and—as Section 3.4 illustrates—are also the metrics most easily corrupted by leakage.

5 Results

5.1 Walk-forward performance

Table 1 reports the walk-forward record. Three observations. First, the gradient-boosted model dominates the logistic benchmark on cohort-level error in five of seven windows, decisively so in the lock-in era (1.1 versus 6.1 CPR points at the 2024 split). Second, both models miss the level of the 2020–21 refinancing wave—as did commercial models; the determinants of wave amplitude (origination capacity, appraisal waivers, media effects) are not spanned by loan-level covariates. Third, discrimination *declines* for all models in the lock-in era: with the entire universe out of the money, the residual variation in who prepays is close to idiosyncratic. The neural network, evaluated at three splits, is competitive on discrimination (AUC 0.75/0.65/0.64) but weaker than the trees on cohort error (8.8/4.6/1.5 versus 8.8/2.8/1.1 CPR points) and under-predicts the hazard level; at 2.5 million training rows this is consistent with the tabular-learning literature, and the original deep-learning advantage of Sadhwani et al. (2021) was demonstrated on three orders of magnitude more data. The gradient-boosted model is therefore the engine for the experiments that follow.

Training window ends	Loan-level AUC		Cohort CPR MAE (pts)	
	Logistic	GBM	Logistic	GBM
2018-12	0.66	0.71	11.6	6.4
2019-12	0.75	0.72	12.7	8.8
2020-12	0.72	0.74	8.6	14.6
2021-12	0.61	0.66	4.6	2.8
2022-12	0.52	0.59	2.6	3.8
2023-12	0.57	0.63	5.2	2.6
2024-12	0.59	0.68	6.1	1.1

Table 1: Walk-forward evaluation. Each row: estimation through December of the stated year, evaluation on the subsequent twelve months. Cohort CPR MAE is UPB-weighted across vintage × coupon cohorts against full-population actuals. The 2020–12 test window is the peak of the pandemic refinancing wave.

5.2 The regime break and the re-estimation trap

We compare a model frozen at December 2021 against annual re-estimation, both walked forward over 2022–25 (Figure 1). The frozen model's failure is immediate: in January 2022 it predicts aggregate speeds of approximately 20 CPR against an actual print of 11, an error of nearly 9 CPR points, as it extrapolates refinancing-wave behavior into a repriced market. The subtler result concerns re-estimation. In 2023, the annually re-estimated model—now fitted through December 2022, including the transition—performs *worse* than the frozen model: 3.8 versus 2.3

CPR points of UPB-weighted error. Six months of partially transitioned data teach the model a speed level that is still too fast, while the frozen model's errors, though structural, are stable and partially offsetting. Re-estimation still trails the frozen benchmark in 2024 (2.6 versus 2.1 CPR points) and overtakes it only in 2025 (1.1 versus 2.4). The governance implication: after a structural break, immediate recalibration on the transition sample can be value-destroying; in this sample, roughly two and a half years of post-break data were required before re-estimation dominated the frozen legacy model. This is consistent with the documented 2023–25 whipsaw in commercial model performance (Katz and Li, 2024).

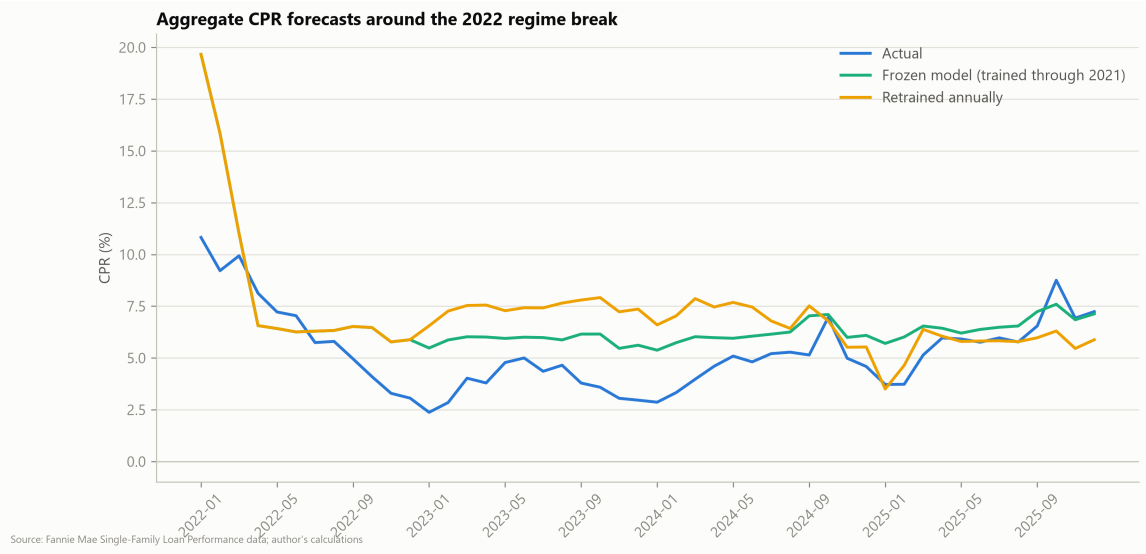

Figure 1: Aggregate UPB-weighted CPR, 2022–2025: actuals versus a model frozen at December 2021 and a model re-estimated each year-end. The two model series coincide during 2022 by construction. The frozen model over-predicts by ≈ 9 CPR points at the January 2022 onset; the re-estimated model over-predicts throughout 2023.

5.3 The learned S-curve, by regime

We estimate the gradient-boosted model separately on each regime—pre-pandemic (2018-01 to 2019-12), refinancing wave (2020-04 to 2021-12), and lock-in (2022-07 to 2025-12)—and trace each fitted model's partial-dependence function of the refinancing incentive on a *common* reference population of 200,000 lock-in-era loan-months. Holding the population fixed ensures that differences between curves reflect learned behavior rather than composition. Figure 2 and Table 2 report the results.

Regime	CPR at −200bp	CPR at 0	CPR at +150bp	Elasticity (pts/100bp)
Pre-pandemic 2018–19	8.8	9.9	13.2	2.2
Refinancing wave 2020–21	20.4	23.9	35.1	7.5
Lock-in 2022–25	8.9	11.7	30.5	12.6

Table 2: S-curve landmarks by regime on the common reference population. Elasticity is the average slope between 0 and +150 basis points of incentive. All entries are partial-dependence values: counterfactual model responses obtained by setting the incentive to the stated level for every observation in the fixed reference population. They are not realized cohort speeds and should not be compared directly to observed cohort CPRs (see text).

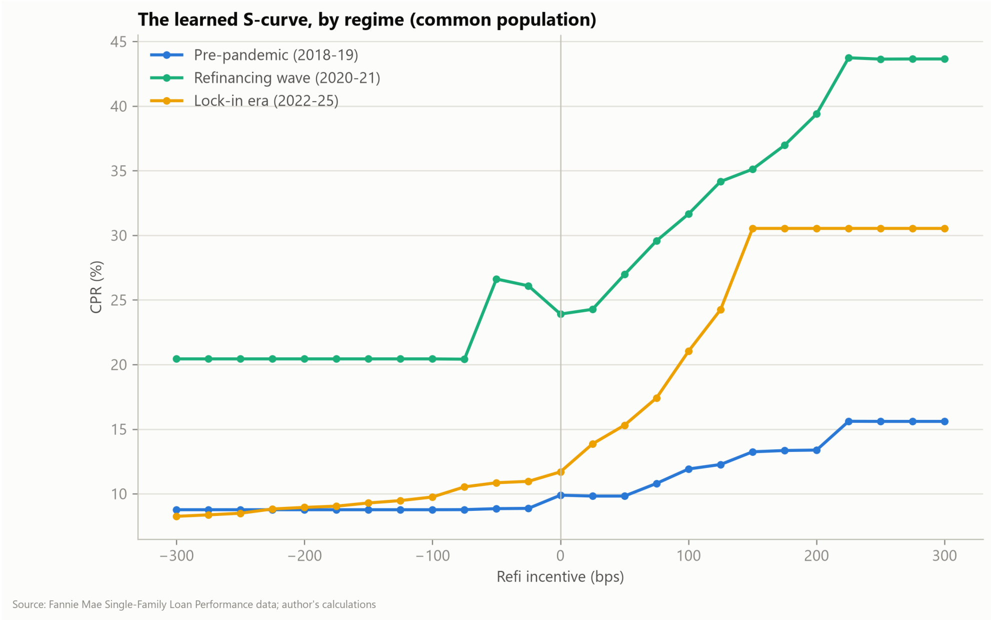

Figure 2: Model-implied S-curves (partial dependence of annualized speed on refinancing incentive) by estimation regime, evaluated on a common lock-in-era reference population.

The central finding is a rotation, not a flattening. The refinancing limb of the lock-in-era S-curve is markedly steeper than its pre-pandemic counterpart: 30.5 versus 13.2 CPR at +150 basis points, an elasticity through the cusp of 12.6 versus 2.2 CPR points per 100 basis points. Borrowers who reach the money in the lock-in era refinance faster than any pre-2020 cohort would have—consistent with the 15-month average age at refinancing observed in the 2024 episode and with the efficiency of modern origination channels. At the deep-discount end, the lock-in model's implied turnover on the common population is close to pre-pandemic levels; the observed 2–4 CPR prints of deeply out-of-the-money cohorts are reproduced through the prevailing-rate-level covariate—the lock-in channel of Batzer et al. (2024)—rather than through a degraded incentive response.

Two remarks prevent misreading of Table 2. First, partial-dependence values at a given incentive are averages over the *entire* reference population with only the incentive altered; all other covariates, including burnout, mark-to-market LTV, and the prevailing-rate level, retain their observed values. The population composition at, say, −200 basis points in the table therefore never occurs in any realized cohort, and the tabulated 8.9 CPR for the lock-in model at −200 basis points is not comparable to the 2–4 CPR prints of 2021-vintage 2.5% coupons: those cohorts sit at −300 to −450 basis points of incentive—at or beyond the left edge of the evaluated grid, where the curves are lower—and their suppressed turnover loads on the rate-level covariate rather than on the incentive itself. Second, because the reference population is fixed across regimes, *differences* between curves are identified as differences in the learned response functions; *levels* inherit the reference population's covariate mix and should be read only relative to one another. SATO-conditioned partial dependence (replication materials) confirms the composition mechanism of Landy (2024): within every regime, higher-SATO tertiles trace steeper curves. The decomposition thus reconciles the composition view with the observation of Katz and Li (2024) that late-2025 realized S-curves exceed pandemic-era steepness: both are features of the same estimated function.

Attribution analysis (mean absolute SHAP values; replication materials) tells the same story at the level of drivers: pre-pandemic prepayment is dominated by burnout and seasoning; the wave by the incentive; and in the lock-in era all drivers compress, consistent with the decline in achievable discrimination.

5.4 Scenario read-through for the current coupon stack

Finally, we apply the full-sample model to the December 2025 cross-section of active 2023–24 vintage 6.0–7.0% coupons under parallel shifts of the mortgage rate, holding burnout and house prices fixed (a deliberately transparent one-month-hazard exercise, annualized). Table 3 reports projected speeds.

Cohort	Unchanged	−50bp	−100bp	−150bp
2023 6.0s	6.9	12.9	20.7	25.9
2023 6.5s	13.1	22.5	29.1	36.0
2023 7.0s	25.4	34.4	43.7	52.0
2024 6.0s	6.5	12.1	19.4	26.1
2024 6.5s	13.3	23.0	32.4	42.1
2024 7.0s	25.5	35.5	43.7	52.2

Table 3: Model-projected CPR for the December 2025 cross-section of 2023–24 vintage cohorts under parallel mortgage-rate shifts.

The 7.0% coupons are already fast at unchanged rates (≈25 CPR) and reach refinancing-wave speeds (above 50 CPR) under a 150 basis point rally; the 6.5s approximately triple over the same interval and constitute the cuspiest segment. This is the model-implied counterpart of the record negative convexity measured in the current-coupon market in late 2025 (Katz and Li, 2024), localized to specific vintage-coupon cohorts.

6 Limitations

Our sample is Fannie Mae collateral only; Ginnie Mae collateral—whose VA segment exhibited the fastest and most policy-sensitive speeds of the 2024–25 episodes—is outside the dataset. The pre-pandemic regime is observed for only two years (vintages 2018+). The scenario analysis is a one-month hazard annualized with burnout and house prices held fixed; it understates path dependence in a sustained rally and is not a valuation (OAS) model. The level of the 2020–21 wave is under-predicted by all specifications, ours and commercial alike, reflecting absent capacity and media-effect covariates. Partial-dependence curves are counterfactuals on a fixed population, not cohort forecasts. Finally, monthly PMMS timing and the publication lag of quarterly house-price indices introduce weeks-level measurement slack, disclosed here and in the replication materials.

7 Conclusion

Using open data and commodity hardware, we provide a public model-side account of the 2022 mortgage prepayment regime break. The prepayment function did not merely shift: its refinancing limb steepened while its turnover component became governed by the prevailing-rate level, and the transition itself created a two-year window in which naive model re-estimation was counterproductive. For practitioners, the results argue for explicit regime-awareness in model governance—freezing, blending, or regime-conditioning rather than reflexive recalibration—and for close attention to the steep embedded optionality of the 2023–24 high-coupon stack. For researchers, the removal-record leakage taxonomy of Section 3.4 is a cautionary catalogue for anyone estimating event models on the public agency datasets: an out-of-sample AUC of 0.9999 is not a discovery.

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A Robustness

This appendix reports four checks on the empirical machinery. Unless stated otherwise, they use the gradient-boosted model estimated through December 2024 and evaluated on 2025, the split with the largest training window.

A.1 Calibration

Figure 3 plots, by decile of predicted hazard, the mean predicted monthly hazard against the design-weighted empirical event rate on the held-out year. The middle deciles lie on the diagonal; the model modestly under-predicts in the slowest deciles (e.g., 0.12% predicted versus 0.16% realized in the first decile) and over-predicts in the fastest decile (2.91% versus 2.40%). Fitted hazards are thus usable as levels, not merely as rankings, with mild attenuation at the fast tail.

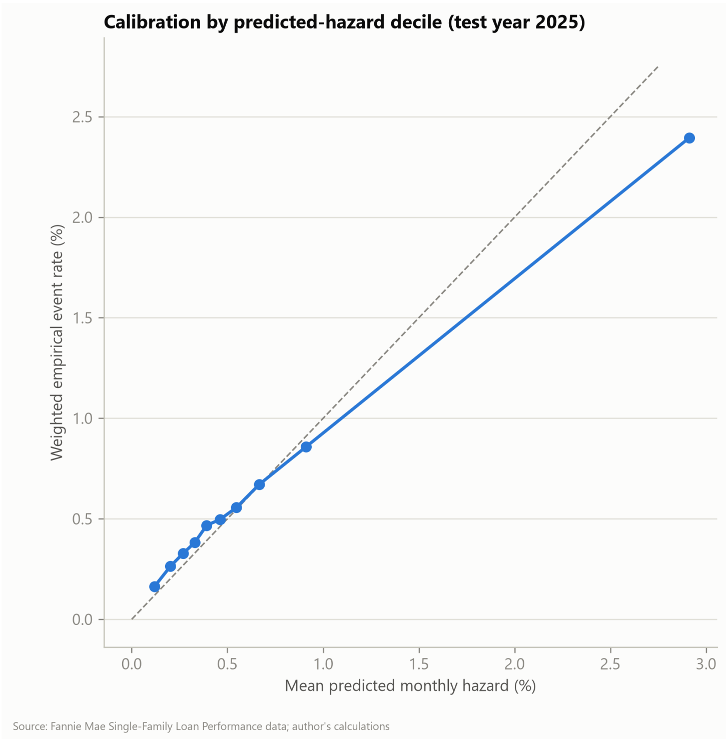

Figure 3: Reliability diagram on the 2025 test year: mean predicted monthly hazard versus design-weighted empirical event rate, by predicted-hazard decile. Dashed line is the 45-degree diagonal.

A.2 Cohort-level fit

Figure 4 plots predicted against actual CPR for the 754 vintage $\times$ coupon cohort-months (with beginning balance above \$1bn) of the 2025 test year; marker area is proportional to cohort balance. The UPB-weighted mean absolute error is 1.09 CPR points, with no visible systematic bias across the speed range.

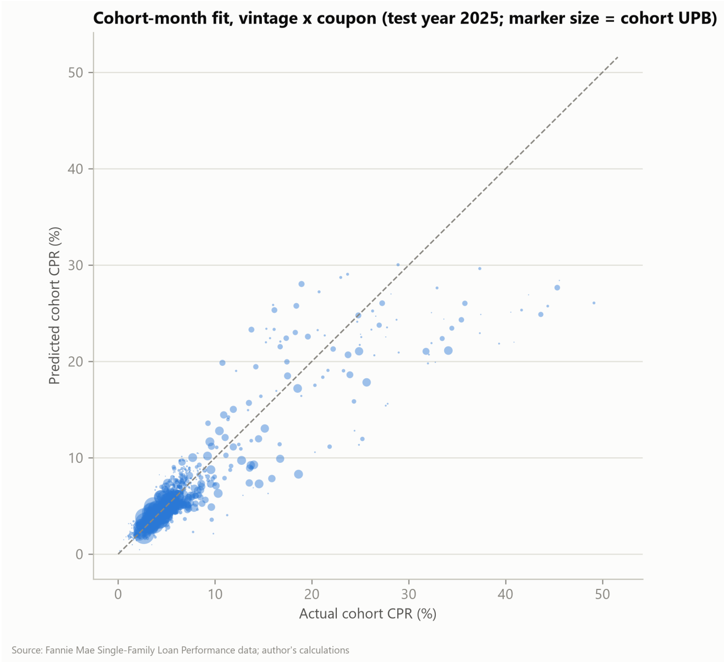

Figure 4: Predicted versus actual cohort CPR, 2025 test year, vintage × coupon cohort-months with beginning UPB above $1bn. Marker area proportional to cohort balance.

A.3 Sensitivity to the non-event sampling rate

The estimation panel retains non-event loan-months with probability $\kappa = 5\%$ (Section 4.1). Re-estimating the December 2024 model after deterministically halving the retained non-events (an effective κ of 2.5%, weights doubled accordingly) leaves results essentially unchanged (Table 4), consistent with the Horvitz–Thompson argument: the downsampling rate affects efficiency, not identification.

Panel	Training rows	AUC (2025)	Cohort CPR MAE (pts)
Baseline, $\kappa = 5\%$	2,037,093	0.691	0.86
Thinned, $\kappa = 2.5\%$	1,206,248	0.690	0.88

Table 4: Sensitivity of the December 2024 model to the non-event retention rate. Evaluation on all 2025 months.

A.4 S-curve robustness across SATO buckets

Figure 5 repeats the partial-dependence analysis of Section 5.3 within tertiles of spread at origination. The regime ordering of Section 5.3 is preserved within every tertile, and within every regime higher-SATO tertiles trace steeper curves — the composition mechanism emphasized by Landy (2024) — confirming that the S-curve rotation is not an artifact of pooling across SATO.

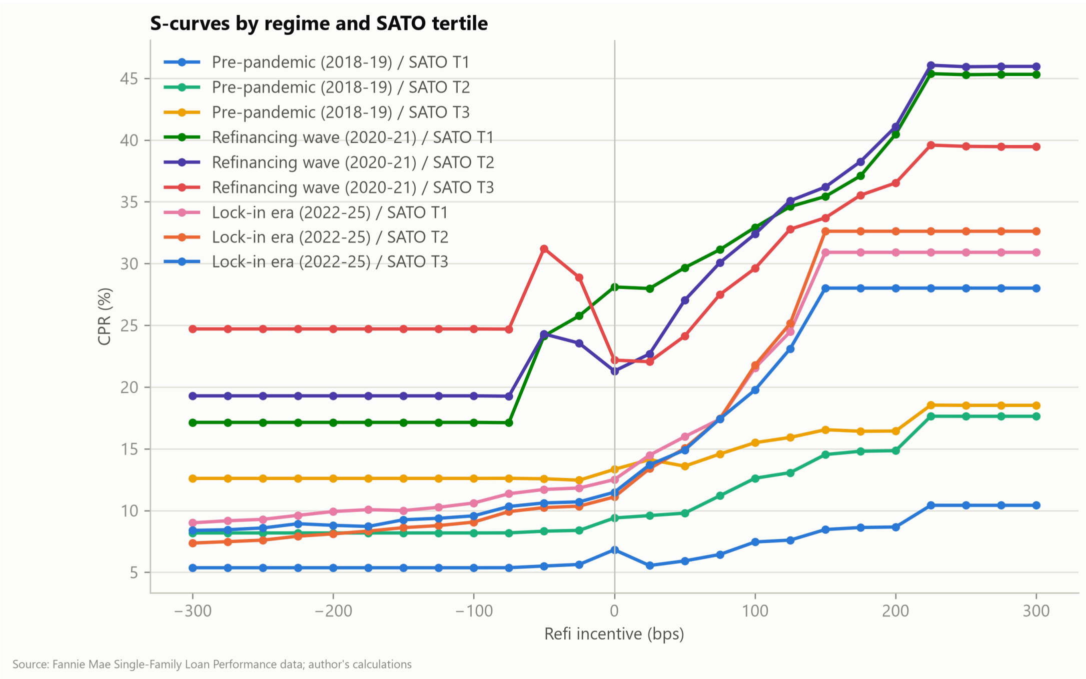

Figure 5: Model-implied S-curves by regime and SATO tertile on the common reference population.